Does In-Domain Masked Autoencoding Overtake ImageNet Initialisation Under Label Scarcity? Two Pre-Registered Studies Across Four Visual Domains

Namrata Pathak
M.Tech in Artificial Intelligence and Data Science
KIIT University, Bhubaneswar, India
namratapathak0410@gmail.com

***Abstract*—For label-poor domains it is often suggested to skip ImageNet and pretrain on the domain's own unlabelled images. We tested this for masked autoencoders (MAE) and a small Vision Transformer (ViT): as labels grow, when does a ViT pretrained with in-domain MAE beat the same ViT started from ImageNet-1k weights, and does distance from ImageNet tell us in advance? Two pre-registered studies covered four domains (plant disease, satellite, objects, animals), leakage-safe splits, label budgets from 5 to 100 % and three seeds on one laptop GPU. In Study 1 (240 runs) the ImageNet ViT won all 20 dataset–budget cells. The gap closed steadily with more labels but never reversed (on PlantVillage from -0.239 to -0.020 macro-F1), and none of four domain distances predicted it (exact $p \geq 0.33$). Study 2 (405 runs) removed three possible handicaps. A four times larger unlabelled pool did not improve MAE (all differences within ± 0.02), a wider learning-rate grid left ImageNet ahead in all 20 cells, and ResNet-34 behaved like ResNet-18. Continuing MAE from the ImageNet weights was never significantly better than ImageNet alone and was significantly worse in 10 of 20 cells. At this scale, ImageNet initialisation is the safer default.**

***Index Terms*—self-supervised learning, masked autoencoders, vision transformers, transfer learning, label efficiency, domain distance, calibration, plant disease**

I. Introduction

Outside everyday photography, labels are usually the costly part of a vision project. Anyone can photograph a tomato leaf, but naming its disease takes a plant pathologist, and satellite or medical images have the same problem. The usual remedy is to start from weights learned on ImageNet [1], 1.28 million labelled photographs of everyday objects.

Self-supervised learning offers another route. A masked autoencoder (MAE) [2] hides most of an image and learns to fill the gaps back in, so it needs no labels and can be trained on exactly the images one cares about. The trade-off is plenty of knowledge from the wrong visual world against a little from the right one. MAE on small target datasets has been reported to help ViTs [3]–[6], and in-domain contrastive pretraining can rival ImageNet in some settings [7]. We could not find an answer to the practical question of *at what label budget* the in-domain route starts to pay off, or whether this can be read off from how unlike ImageNet a domain is, a link that Ericsson et al. [8] leave open.

We address this with two pre-registered studies. Our contributions are: (1) a comparison of four starting points on four domains, in which the three ViTs share one architecture and differ only in their initial weights; (2) a crossover analysis over five label budgets, showing that in-domain MAE never overtakes ImageNet although the gap shrinks in every domain; (3) four domain distances measured before training, which disagree with each other and do not predict the gap; (4) a follow-up that removes three possible handicaps of MAE and tests a hybrid that continues MAE from ImageNet weights, none of which changes the conclusion; and (5) evidence that accuracy rankings replicate across domains while calibration rankings do not. Both studies were committed to a public repository before their first training run and took about 36 h on a laptop GPU in total.

II. Related Work

Transfer from ImageNet. Stronger ImageNet models tend to transfer better [9], yet for medical images much of what ImageNet offers turns out to be unnecessary [10]. ViTs [11] have no built-in locality, unlike CNNs [12], and depend heavily on pretraining; DeiT [13] showed they can be trained on ImageNet-1k alone.

In-domain and continued pretraining. The value of self-supervised features depends on the target domain [8], and both the amount and the domain of pretraining data matter for contrastive models [7]. Masked pretraining on the target data alone can be competitive [3], [4] and reduces the data needs of ViTs [5], [6]; the diversity of the pretraining set also plays a part [14]. Continuing pretraining on in-domain data is common in language modelling [15], and Reed et al. [16] report gains when self-supervised training starts from ImageNet weights. For plants, CNNs reach high accuracy on PlantVillage [17], [18], and self-supervised transformers have been trained on plant images [19]. None of these locates the label budget at which in-domain MAE overtakes ImageNet across several domains.

Distances and calibration. FID [20] compares feature distributions, and OTDD [21] adds the labels. Temperature scaling [22] is the standard fix for overconfidence, and calibration has been revisited for newer models [23] and for small medical datasets [24]. Since results on one benchmark often fail to carry over [25], we use several domains and rank statistics [26].

III. Method

A. Arms

All arms are fine-tuned on the same labelled subsets (Fig. 1). Study 1 has four: **A**, a ResNet-18 [12] from random weights (11.2 M parameters); **C**, a ViT-Tiny in the DeiT-Tiny layout [13] (12 blocks, width 192, 5.5 M parameters) with supervised ImageNet-1k weights; **D**, the same ViT from random weights; and **E**, the same ViT initialised by an MAE trained on the domain's own training images without labels. As C, D and E share one architecture, their differences come only from the starting weights. E−C is the main contrast; E−D shows what MAE adds, and E−A compares it with a CNN. Study 2 adds **E2**, which is E with a four times larger unlabelled pool; **F**, which takes C's ImageNet weights, continues MAE on that pool and is then fine-tuned; and **A2**, a ResNet-34 (21.3 M parameters). Model sizes and the 160 px resolution were chosen so that each study fits within 48 h on one RTX 4050 Laptop GPU (6 GB).

B. Data, splits and label budgets

Each domain is one dataset (Table I): PlantVillage [17] (leaf diseases), EuroSAT RGB [27] (land cover), CIFAR-10 [28] (objects) and Oxford-IIIT Pets [29] (cat and dog breeds). We draw 7,000 images per dataset, stratified by class, resize the short side to 160 px and crop the centre. Near-copies, such as repeated shots of one leaf, would inflate test scores if they straddled a split, so we hash each image (256-bit difference hash), group images whose hashes differ in at most 10 bits, and keep every group inside one split of a 70/15/15 `StratifiedGroupKFold` split. The label budgets are 5, 10, 20, 50 and 100 % of each class's training images (at least one per class), i.e. about 250 to 5,000 labels. Smaller budgets are subsets of larger ones, and every arm sees the same subset for a given seed (0, 1 or 2), which also fixes data order, new-layer initialisation and E's MAE run.

C. Distance from ImageNet

Images are embedded with C's frozen ImageNet ViT (192-d class token) and compared with ImageNetV2 (matched frequency, 10,000 images) [30] as a stand-in for ImageNet. We measure (i) FID [20]; (ii) OTDD [21], whose matching cost also reflects class similarity (diagonal Gaussian per class, 64 PCA dimensions, 1,500 samples per side, POT solver [31]); (iii) ImageNet confidence, the mean top-class probability of the ImageNet classifier on the domain (0.458 on ImageNetV2 itself); and (iv) the validation accuracy of a linear probe on the frozen features. Higher FID or OTDD and lower confidence or probe accuracy mean "farther away".

D. MAE pretraining

The MAE encoder is the same ViT-Tiny, so its weights load directly into the fine-tuned model. It hides 75 % of the 100 patches, and a small decoder (2 blocks, width 128) predicts the normalised pixels of the hidden ones. We use batch 256, AdamW ($\beta_2 = 0.95$), learning rate 1.5×10^{-4}, weight decay 0.05, cosine decay, random resized crops and flips, with sine–cosine position embeddings at the start. For E the pool is the ∼5,000 training images, seen for 800 epochs (15,200 steps, ∼21.5 min); each seed gets its own MAE. In Study 2 the pool holds 20,000 images: the training images plus 15,000 more drawn from *outside* the 7,000-image sample, so validation and test images cannot enter. Extras whose hash is within 10 bits of a validation or test image, or of a flipped copy, are dropped as well (227 on EuroSAT, none elsewhere). E2 keeps E's 15,200 steps, so only the data changes. F uses the same schedule from the ImageNet weights, keeps their position embeddings and lowers the learning rate to 5×10^{-5} to avoid erasing them; this value was fixed in advance, not tuned. Oxford Pets has no further images, so it has no E2 and its F uses the 5,000 training images.

E. Fine-tuning and statistics

All arms follow Table II. Validation macro-F1 is checked 25 times per run; training stops after 8 checks without gain and the best weights are restored before the single test evaluation. Pretrained ViTs use layer-wise learning-rate decay of 0.75 [32]. The learning rate per dataset and arm is chosen on validation macro-F1 at the 20 % budget with seed 0 and reused everywhere. Study 2 widens the grid to five values and retunes every arm; old runs are reused where the chosen rate is unchanged and repeated where it changed.

The main metric is test macro-F1; we also report the expected calibration error (ECE, 15 bins [33]) before and after temperature scaling fitted on validation data [22]. Gaps between arms are paired by seed and given with 95 % t-intervals. The crossover budget is the sign change of the mean gap on a log-budget axis; without one it is reported as censored (>100 %), and it counts only if the intervals exclude zero on both sides. Each dataset's gap is summarised by its average over log-budget, which we relate to each distance by Spearman's ρ with an exact permutation p and a leave-one-dataset-out prediction. With four datasets no two-sided p can fall below 0.083, so this analysis was declared exploratory. Ranking agreement across datasets uses Kendall's W [34] and the Friedman test.

Pre-registration. Hypotheses H1–H5, the protocol and the analysis plan of Study 1 were committed before its first run; one amendment, also before training, raised MAE from 400 to 800 epochs after a speed test. Hypotheses H6–H11 were committed after Study 1 ended and before any Study 2 code ran (Table V).

IV. Study 1: Results

All 240 runs finished in 14.6 h (9.3 h fine-tuning, 1.0 h learning-rate selection, 4.3 h MAE), using at most 3.4 GB of

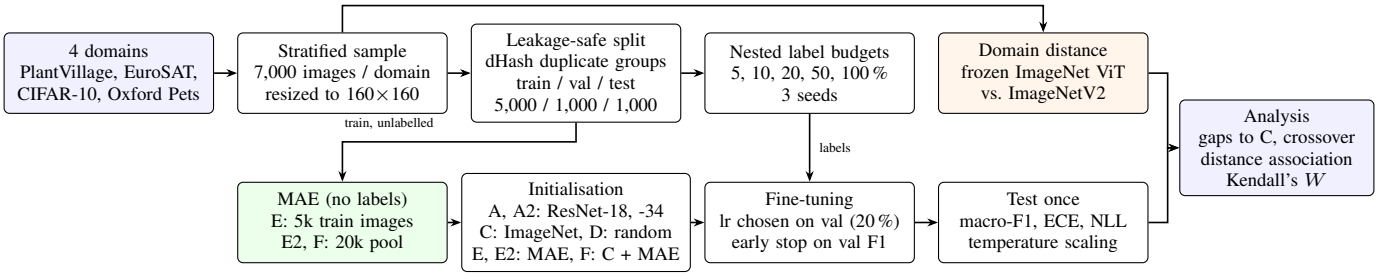


Fig. 1. Pipeline, run separately for each domain. Study 1 uses arms A, C, D and E; Study 2 adds A2, E2 and F. Only MAE pretraining sees images before fine-tuning, and never validation or test images. Each run is evaluated on the test set once.

TABLE I
DATASETS, LEAKAGE-SAFE SPLITS AND DISTANCES TO IMAGENET, MEASURED BEFORE TRAINING. THE ARROW SHOWS WHICH DIRECTION MEANS "FARTHER FROM IMAGENET".

Dataset	Domain	Classes	Native px	Train/Val/Test	Dup. groups	FID ↑	OTDD ↑	ImageNet conf. ↓	Linear probe ↓
PlantVillage	plant	38	256	4999/1000/1001	0	433	30.6	0.30	0.90
EuroSAT	satellite	10	64	5000/1000/1000	1	318	26.4	0.14	0.86
CIFAR-10	objects	10	32	5000/1000/1000	0	267	28.3	0.29	0.83
Oxford Pets	animals	37	var.	5000/1000/1000	34	273	29.3	0.51	0.86

TABLE II
FINE-TUNING PROTOCOL, SHARED BY ALL ARMS.

Optimiser	AdamW [35], weight decay 0.05
Batch / precision	128 / bf16, gradient clipping 1.0
Schedule	5 % warm-up, cosine decay
Steps	$\mathrm{clip}(100\lceil n/128\rceil, 600, 4000)$
Augmentation	resized crop 0.35–1, flip, colour ±20 %
ViT details	drop-path 0.1, average pooling, LLRD 0.75
LR grid, scratch	S1: $\{0.5, 1, 2\}$, S2: $\{0.25, \ldots, 4\} \times 10^{-3}$
LR grid, pretrained	S1: $\{0.25, 0.5, 1\}$, S2: $\{0.125, \ldots, 2\} \times 10^{-3}$

TABLE III
STUDY 1 TEST MACRO-F1 (MEAN ± S.D. OVER 3 SEEDS). BOLD: BEST ARM. LAST COLUMN: PAIRED GAP E−C; *95 % CI EXCLUDES 0.

	%	A: CNN scr.	C: ViT IN	D: ViT scr.	E: ViT MAE	E−C
PlantVillage	5	0.548 ±0.019	**0.806 ±0.012**	0.500 ±0.038	0.567 ±0.020	−0.239*
	10	0.697 ±0.033	**0.861 ±0.039**	0.607 ±0.019	0.693 ±0.019	−0.167*
	20	0.810 ±0.016	**0.940 ±0.006**	0.700 ±0.010	0.812 ±0.024	−0.127*
	50	0.930 ±0.007	**0.979 ±0.010**	0.821 ±0.008	0.925 ±0.014	−0.054*
	100	0.970 ±0.003	**0.982 ±0.006**	0.911 ±0.005	0.962 ±0.005	−0.020*
EuroSAT	5	0.764 ±0.016	**0.934 ±0.009**	0.714 ±0.032	0.751 ±0.005	−0.183*
	10	0.813 ±0.016	**0.954 ±0.010**	0.784 ±0.004	0.799 ±0.020	−0.155*
	20	0.851 ±0.014	**0.964 ±0.004**	0.846 ±0.011	0.853 ±0.010	−0.111*
	50	0.913 ±0.010	**0.977 ±0.003**	0.895 ±0.006	0.910 ±0.007	−0.066*
	100	0.949 ±0.006	**0.982 ±0.003**	0.918 ±0.007	0.933 ±0.003	−0.049*
CIFAR-10	5	0.382 ±0.012	**0.868 ±0.020**	0.314 ±0.030	0.425 ±0.017	−0.443*
	10	0.452 ±0.026	**0.887 ±0.006**	0.365 ±0.032	0.506 ±0.004	−0.380*
	20	0.537 ±0.017	**0.904 ±0.008**	0.419 ±0.007	0.581 ±0.014	−0.323*
	50	0.674 ±0.004	**0.926 ±0.008**	0.501 ±0.005	0.672 ±0.005	−0.255*
	100	0.746 ±0.006	**0.932 ±0.010**	0.565 ±0.011	0.736 ±0.018	−0.196*
Oxford Pets	5	0.117 ±0.010	**0.771 ±0.008**	0.073 ±0.006	0.151 ±0.033	−0.620*
	10	0.179 ±0.013	**0.815 ±0.008**	0.102 ±0.013	0.210 ±0.020	−0.604*
	20	0.283 ±0.014	**0.853 ±0.007**	0.122 ±0.007	0.305 ±0.009	−0.548*
	50	0.466 ±0.001	**0.880 ±0.006**	0.196 ±0.004	0.502 ±0.018	−0.378*
	100	0.659 ±0.015	**0.890 ±0.009**	0.286 ±0.014	0.638 ±0.017	−0.252*

GPU memory.

No crossover. C had the best mean macro-F1 in all 20 dataset–budget cells (Fig. 2, Table III). E−C was negative everywhere and its 95 % interval excluded zero in every cell, so the crossover is censored at >100 % for all datasets. H1 holds only in a weak form: the gap closes with more labels in every domain but never flips. On PlantVillage it shrank from -0.239 at 5 % to -0.020 at 100 % (CI $[-0.037, -0.003]$) and on EuroSAT from -0.183 to -0.049, while CIFAR-10 (-0.443 to -0.196) and Oxford Pets (-0.620 to -0.252) kept a wide gap.

Distance does not predict the gap. Averaged over budgets, the gap was much smaller for PlantVillage (-0.118) and EuroSAT (-0.110) than for CIFAR-10 (-0.317) and Pets (-0.489), whose objects and animals ImageNet covers well; on Pets the ImageNet classifier is even more confident (0.51) than on ImageNetV2 (0.46). Yet the four distances disagree on which domain is far away (Table I): FID puts PlantVillage farthest, confidence puts EuroSAT farthest and OTDD puts EuroSAT *closest*. FID ($\rho = +0.6$) and confidence ($\rho = -0.8$) point the way H2 expected, OTDD the other way ($\rho = -0.4$), and the linear probe not at all ($\rho = 0$). No exact p is below 0.33, and only confidence predicts a held-out dataset better than the mean (error 0.124 against 0.193). H2 is not supported.

MAE versus scratch. E beat the scratch ViT in all 20 cells, significantly in 13 (H3 mostly supported). The gain was largest on Pets ($+0.353$ at 100 %, 0.638 against 0.286) and CIFAR-10 ($+0.111$ to $+0.171$), and small and never significant on EuroSAT ($+0.007$ to $+0.037$), where the scratch ViT was already good. Against the scratch ResNet-18, E stayed between -0.020 and $+0.055$, significant in only 2 of 20 cells. At this scale MAE gives a ViT about what its convolutional design gives a ResNet.

Rankings and calibration. The order of the arms by macro-F1 (C, then E and A, then D) was nearly identical across domains: Kendall's W was 0.90–1.00 at every budget (Friedman

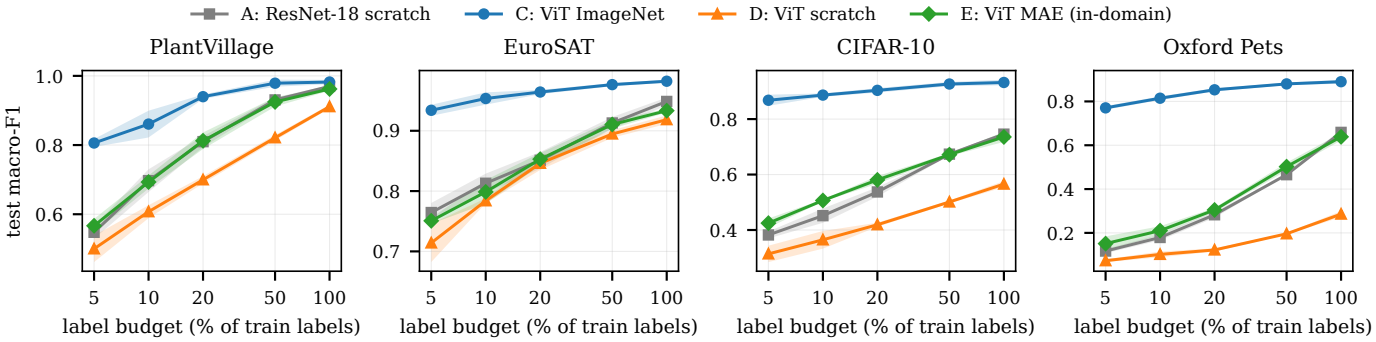


Fig. 2. Study 1: test macro-F1 against the label budget (mean of three seeds, band = ±1 s.d.). The ImageNet ViT (C) is on top everywhere. In-domain MAE (E) sits above the scratch ViT (D) and close to the scratch ResNet-18 (A).

$p \leq 0.013$), so H4 is rejected. Before temperature scaling C also had the lowest ECE in 17 of 20 cells (mean 0.052, against 0.119 for E, 0.154 for D and 0.171 for A). Scaling brought every arm to 0.024–0.032 on average, after which the ranking no longer agreed across datasets ($W = 0.05$–0.53, all $p \geq 0.098$), and the scratch ViT was the best calibrated in 7 cells. H5 is partly supported.

V. Study 2: Was Study 1 Unfair to MAE?

A negative result raises the question of whether the losing method had a fair chance. Study 2 tests the three objections we found most convincing and adds the hybrid F. It used 405 runs, reusing unchanged Study 1 cells, and took about 21 h; the GPU ran at full load but throttled at ∼89 °C, so an MAE run took 22.8 instead of 21.5 min. Fig. 3 and Table IV summarise the results.

Wider learning-rate grid. With five candidate rates the choice changed for 9 of the 16 Study 1 dataset–arm pairs, and those 135 runs were repeated. Nothing important moved: C still won all 20 cells among A, C, D and E (H11 supported), and E−C stayed negative everywhere, significantly in 19 cells. The closest case is again PlantVillage at full labels, now −0.012 instead of −0.020.

Four times more unlabelled images. If MAE had simply been short of data, a larger pool should have helped. It did not: E2 and E differed by at most 0.02 in any cell, E2 led in 8 of 15 and no difference was significant (H6 not supported). E2 stayed below C in all 15 cells, 14 significantly, and came no closer than 0.014 (PlantVillage, 100 %; H7 not supported). Even the reconstruction loss barely changed (0.407 against 0.409 on PlantVillage, 0.266 against 0.268 on CIFAR-10).

ImageNet first, then MAE. The hybrid F beat C in only 3 of 20 cells, by at most 0.006 and never significantly, and was significantly worse in 10 (H8 not supported). The harm is concentrated at small budgets: at 5 % F trailed C by 0.351 on CIFAR-10 and 0.542 on Pets, most of ImageNet's original lead over in-domain MAE. With all labels it came back to within 0.04 of C everywhere. F was well ahead of E2 (0.021 to 0.210, significant in 14 of 15 cells; H9 supported), but that edge comes from the ImageNet start. F also stopped early in only 17 % of its runs, close to E (20 %) and far from C (67 %).

A larger ResNet. ResNet-34 and ResNet-18 were within 0.03 in 18 of 20 cells; the one significant difference was CIFAR-10 at 100 % (+0.042). The verdict of MAE against a scratch CNN changed in a single cell (H10 partly supported). Mean ECE before scaling was 0.052 for C, 0.116 for E2, 0.146 for F, 0.151 for E and 0.176 for ResNet-34; after scaling all arms fell between 0.024 and 0.033, as in Study 1.

VI. Discussion

Why ImageNet keeps winning. ImageNet-1k has about 250 times more images than Study 1's pool and 64 times more than Study 2's, and its labels force the network to learn shapes, parts and textures that prove useful even for leaves and satellite tiles. MAE learns by repainting pixels. Its reconstructions (Fig. 4) recover the layout, such as a leaf's outline and colour, but blur fine detail, including the lesions that separate many PlantVillage classes. With enough labels, fine-tuning can learn that detail, which fits a gap that shrinks with the budget. That four times more images left both the reconstruction loss and the accuracy unchanged suggests the limit lies in what pixel reconstruction teaches at this model size rather than in how many images it sees. MAE is usually trained on far larger collections [2], so our result means "four times more does not help", not "more can never help".

Why continued MAE hurt. Continuing with MAE seems to pull the ImageNet network away from the class-level features that made C strong and towards features suited to repainting pixels. With few labels there is too little signal to pull it back, so the damage is largest at 5–20 % and almost gone at 100 %. Reed et al. [16] saw the opposite, but they started from self-supervised weights and used contrastive objectives. Our learning rate for F was also a fixed guess, and F ended with a higher reconstruction loss than MAE from scratch (0.456 against 0.407 on PlantVillage), so a gentler or shorter schedule may behave differently. This simple recipe, at least, is not a free improvement.

Where in-domain MAE comes closest. The gap was smallest for PlantVillage and EuroSAT, the domains least like ImageNet's object photos, as H2 intended, but no distance captured it; one number seems too blunt to forecast a crossover. EuroSAT also had by far the lowest reconstruction loss (0.19,

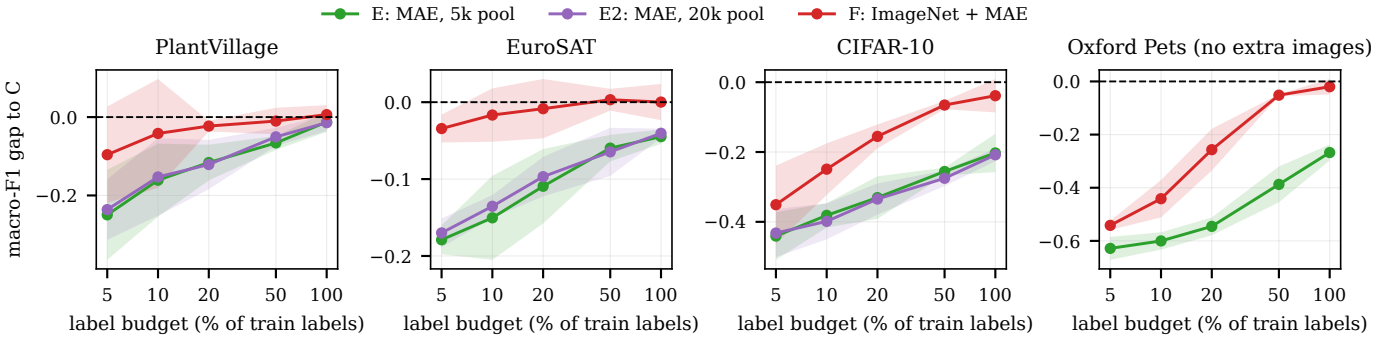


Fig. 3. Study 2: paired macro-F1 gap to the ImageNet ViT (C) for in-domain MAE on 5,000 images (E), on 20,000 images (E2), and for ImageNet weights with continued MAE (F), with 95 % t-intervals over three seeds. E2 lies on top of E. F starts far below C and catches up only when labels are plentiful.

TABLE IV
STUDY 2 PER DATASET, AFTER RETUNING ON THE WIDER LEARNING-RATE GRID. MEAN GAPS ARE AVERAGED OVER THE LOG-BUDGET AXIS (MACRO-F1). "SIG. +/−": BUDGETS WHOSE 95 % CI LIES ENTIRELY ABOVE / BELOW 0. "F>C / SIG. −": BUDGETS WHERE F'S MEAN IS HIGHER THAN C'S / BUDGETS WHERE F IS SIGNIFICANTLY LOWER.

	E−C	E2−E (more unlabelled data)		E2−C	F−C	F>C / sig. −	A2−A (ResNet-34 − 18)
Dataset	mean gap	mean gap	sig. +/−	mean gap	mean gap	(of 5 budgets)	range over budgets
PlantVillage	−0.116	+0.006	0/0	−0.110	−0.029	1/1	−0.015 to +0.009
EuroSAT	−0.106	+0.007	0/0	−0.099	−0.009	2/1	−0.008 to +0.003
CIFAR-10	−0.320	−0.010	0/0	−0.330	−0.162	0/4	−0.002 to +0.042
Oxford Pets	−0.493	–	–	–	−0.250	0/4	−0.011 to +0.034

TABLE V
ALL PRE-REGISTERED HYPOTHESES AND THEIR OUTCOMES. H1–H5 WERE REGISTERED BEFORE STUDY 1, H6–H11 BEFORE STUDY 2.

	Hypothesis	Outcome
H1	C ahead at small budgets; E closes or reverses the gap at larger budgets	**Partly**: the gap closes in all 4 domains, but no crossover up to 100 %
H2	Gap E−C smaller for domains farther from ImageNet	**Not supported**: right direction for 2 of 4 distances, none significant
H3	E > D at every budget in every domain	**Mostly**: 20/20 positive, 13/20 significant, none of them on EuroSAT
H4	The best arm differs across domains and budgets ($W < 1$)	**Rejected**: C best in 20/20 cells, $W = 0.90$–1.00
H5	Ranking by ECE differs from ranking by macro-F1	**Partly**: similar before temperature scaling, different after
H6	A 20,000-image pool improves MAE (E2 > E)	**Not supported**: differences within ±0.02, 8/15 positive, none significant
H7	E2 narrows the gap to C at every budget and overtakes C somewhere	**Not supported**: E2 below C in 15/15 cells (14 significant)
H8	ImageNet + MAE (F) beats C in most cells, more so for distant domains	**Not supported**: F > C in 3/20 (none significant), significantly worse in 10/20
H9	F ≥ E2 on the three datasets with a pool	**Supported**: F ahead by 0.021–0.210, 14/15 significant
H10	ResNet-34 within 0.03 of ResNet-18 in most cells; MAE-vs-CNN verdict unchanged	**Partly**: within 0.03 in 18/20; verdict flips in 1 cell
H11	The wider lr grid does not change the winner among A, C, D, E	**Supported**: C still best in 20/20 cells

against 0.27–0.54 elsewhere) and the smallest gain over a scratch ViT, so a low reconstruction loss is no sign of a useful representation.

Practical advice. With up to twenty thousand unlabelled images and a small ViT, start from ImageNet at every label budget. In domains unlike ImageNet its lead at full labels can be small (about one point on PlantVillage), and in-domain MAE is still a clear step up from training a ViT from scratch. Calibration should be checked per domain, since the accuracy ranking does not carry over to it.

VII. LIMITATIONS

The studies were sized for a laptop GPU. (i) ViT-Tiny, 160 px and at most 20,000 unlabelled images per domain; larger models, resolutions and pools favour MAE [2], so a crossover may exist beyond our range. (ii) Four domains cap any rank correlation at $p \geq 0.083$, and three seeds leave wide intervals for small effects. (iii) Even on the wider grid, 16 of 27 chosen learning rates lie at an end of the grid; since widening it barely changed the results, we do not expect further tuning to reverse them. (iv) One MAE recipe was used, and F's MAE learning rate was not tuned. (v) DeiT weights were trained at 224 px and interpolated to 160 px,

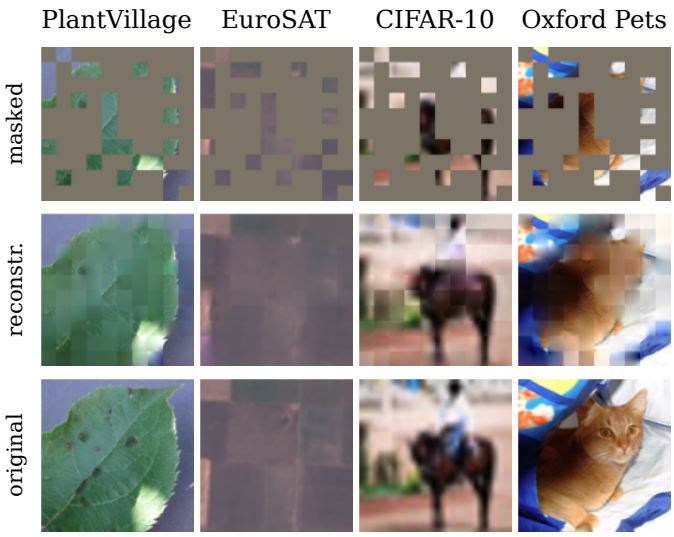


Fig. 4. MAE reconstructions from Study 1 (seed 0, one training image per domain). Top: input with 75 % of the patches hidden; middle: reconstruction; bottom: original. The layout comes back, fine detail such as leaf lesions does not.

and CIFAR-10 and EuroSAT were upsampled. (vi) Hashing cannot catch every near-duplicate, such as a leaf photographed from another angle. (vii) ImageNetV2 replaces ImageNet in the distances, Pets had no extra images for E2, and compute was not equalised between arms.

VIII. Conclusion

Over four domains and two pre-registered studies, in-domain MAE did not overtake ImageNet initialisation of a small ViT anywhere between 5 and 100 % of the labels. ImageNet's lead shrank as labels were added and was smallest for plant-disease and satellite images, but no standard distance predicted where it would be small. Four times more unlabelled data, a wider learning-rate grid and a deeper ResNet baseline did not change this, and continuing MAE from ImageNet weights hurt when labels were scarce. In-domain MAE did reliably improve a ViT trained from scratch and matched a CNN trained from scratch. Much larger unlabelled pools, larger ViTs and more domains are the natural next steps, and would also allow the link between domain distance and the crossover to be tested properly.

Reproducibility

Code, configurations, both pre-registrations, split files, every run's metrics and the logs are available at

https://github.com/namratapathak0410-pixel/Label-Efficient-Representation-Learning-SSL.

Models were built with PyTorch and timm [36]; every figure and table is produced from the result files by `paper/make_figures.py`.

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